

normalized gain over PPO

- cgae-cf – PPO, paired, 95% CI (20 seeds); filled $p < 0.05$
- not significant
- ceiling: headroom PPO leaves, $1 - \text{PPO}$ (per configuration)

single component 1.0
 $N = 2$ 2.0
 $N = 4$ 3.3
hard $N = 4$ 3.1
 $N = 8$ 6.3
multi-site 8.0

K (reward-bearing causal components, measured before training)

